Ernest Ma

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Education

San Jose State University **December 2026**

Bachelor of Science in Computer Engineering | GPA: 3.95 *San Jose, CA*

- **Organizations**: Software and Computer Engineering Society, Spartan Racing: Formula SAE, HKSA
- **Relevant Coursework**: Data Structures and Algorithms, Computer Architecture, Information Security, Microprocessor Design, Operating Systems, Database Systems, Computer Networks

Experience

KLA | Software Engineer Intern Ann Arbor, MI | May 2026 – August 2026

- Accelerated physics-ML training via **PyTorch DDP** + **ClearML** on **Kubernetes**, scaling 3.9× throughput on 4 GPUs (98% efficiency)
- Engineered a **Kubernetes** AI agent **(kagent + MCP + Prometheus)** with 12 custom tools for health scans, alert triage, and reports
- Built a **React/FastAPI** dashboard with **PostgreSQL** and **Prometheus** for **K8s** clusters, **ClearML** experiments, and **VMware**
- Standardized a **LangGraph** template and Kubernetes deployment workflows, reducing a project cycle from weeks to days

KLA | Software Engineer Intern Ann Arbor, MI | May 2025 – August 2025

- Built a troubleshooting agent using **LangGraph**, **FastAPI**, and **Gemma 3 / GPT-4o** to assist field engineers
- Reduced machinery downtime by **80%** by automating parsing of complex system logs and enabling natural language queries
- Engineered a **RAG** pipeline with ChromaDB to accelerate historical log retrieval from hours to sub-second results
- Deployed and configured an **AWS EC2 (Linux)** environment, optimizing for **sub-second** retrieval and **99.9%** uptime

SJSU College of Engineering | Software Engineer Intern San Jose, CA | January 2026 – May 2026

- Architected a **Go/Gin** event service processing registrations and waitlists for **1k+** concurrent users via a MongoDB backend
- Mitigated data races during **10k+ write/sec** spikes by implementing **Redis** concurrency locks and optimized caching strategies
- Orchestrated a **Kafka** event-driven pipeline to decouple API requests from registration processing, ensuring strong scalability

Spartan Racing: Formula SAE | Software Engineer Intern San Jose, CA | January 2026 – May 2026

- Expanded a **C-based** Hardware in Loop injection system for testing using CAN protocols for real-time VCU validation
- Developed a **Python** HIL test suite utilizing state machines to manage complex CAN communication sessions
- Synchronized automated telemetry validation to ensure data consistency during high-speed testing cycles
- Verified Launch Control accuracy by hand-calculating torque outputs against parameterized test cases

Projects

Agent Faultline — *Rust, Python, MCP, PostgreSQL, Docker, OpenTelemetry*

- Built **Rust/Python** MCP benchmarks across 2 **PostgreSQL** environments with **5** deterministic tool-fault types for agent recovery
- Engineered a Rust proxy and scoring pipeline; validated **10K** idempotency and deployment attempts for error detection
- Added scoped stdio/HTTP MCP support; validated safe timeout recovery with a filesystem server under fault injection

PatchBench — *Python, pytest, Docker, YAML*

- Built an executable benchmark to measure whether AI code reviewers detect regressions that pass visible tests
- Designed an 18-task **Python** corpus with 36 correct and regression patches, test isolation, and **Docker-based** replay
- Implemented a CLI reviewer harness for structured findings and deterministic detection, false-positive, and inconclusive scoring

Collab Code — *TypeScript, VS Code Extension API, Liveblocks, Yjs*

- Built a real-time collaborative IDE extension using **Yjs** CRDTs and **Liveblocks** for multi-user editing and room sessions
- Implemented synchronized cursors, selections, and presence indicators with document binding and local-origin filtering
- Created account-less shared-key onboarding and automated VSIX packaging for seamless cross-editor distribution

SJSU Safeline — *Nemotron, FastAPI, React / Vite, LangChain, SQLite*

- Built an agentic AI assistant using **NVIDIA Nemotron** and **FastAPI** to automate crisis triaging and personalized resource routing
- Architected a reasoning pipeline with **LangChain** function-calling for urgency assessment and real-time resource retrieval
- Implemented a multi-stage workflow to provide 24/7 student support through generating personalized outreach scripts

Technical Skills

Languages: Go, Python, C++, C, SQL, C#, Java, Dart, Verilog, x86 Assembly
Frameworks & Libraries: FastAPI, React, PyTorch, LangGraph, LangChain, Flutter, Yjs (CRDTs)
Infrastructure & Cloud: Kubernetes, Docker, Helm, AWS (EC2), ClearML, Prometheus, Kafka, Linux
Databases: PostgreSQL, MongoDB, Redis, ChromaDB, Firebase